mechanics radically challenges this view. *Two things can be separated by an enormous amount of space and yet not have a fully independent existence.* A quantum connection can unite them, making the properties of each contingent on the properties of the other. Space does not distinguish such entangled objects. Space cannot overcome their interconnection. Space, even a huge amount of space, does not weaken their quantum mechanical interdependence.

Some people have interpreted this as telling us that "everything is connected to everything else" or that "quantum mechanics entangles us all in one universal whole." After all, the reasoning goes, at the big bang everything emerged from one place since, we believe, all places we now think of as different were the same place way back in the beginning. And since, like the two photons emerging from the same calcium atom, everything emerged from the same something in the beginning, everything should be quantum mechanically entangled with everything else.

While I like the sentiment, such gushy talk is loose and overstated. The quantum connections between the two photons emerging from the calcium atom are there, certainly, but they are extremely delicate. When Aspect and others carry out their experiments, it is crucial that the photons be allowed to travel absolutely unimpeded from their source to the detectors. Should they be jostled by stray particles or bump into pieces of equipment before reaching one of the detectors, the quantum connection between the photons will become monumentally more difficult to identify. Rather than looking for correlations in the properties of two photons, one would now need to look for a complex pattern of correlations involving the photons and everything else they may have bumped into. And as all these particles go their ways, bumping and jostling yet other particles, the quantum entanglement would become so spread out through these interactions with the environment that it would become virtually impossible to detect. For all intents and purposes, the original entanglement between the photons would have been erased.

Nevertheless, it is truly amazing that these connections do exist, and that in carefully arranged laboratory conditions they can be directly observed over significant distances. They show us, fundamentally, that space is not what we once thought it was.